THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 9: **Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw a neat figure for every question of Sections B, C and D. Take $\sqrt{3} = 1.732$ where needed.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ladder leaning against a wall makes 60° with the ground. Its foot is 2.5 m from the wall. The length of the ladder is: 1

(a) 5 m **(b)** $2.5\sqrt{3}$ m **(c)** $5\sqrt{3}$ m **(d)** 2.5 m

2. A 6 m high pole casts a shadow $2\sqrt{3}$ m long. The sun's elevation is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

3. From the top of a 50 m tower, the angle of depression of a car is 45°. The distance of the car from the tower is: 1

(a) 50 m **(b)** $50\sqrt{3}$ m **(c)** 25 m **(d)** $\frac{50}{\sqrt{3}}$ m

4. The angle of elevation of the top of a tower from a point 100 m away is 60°. Its height is: 1

(a) $100\sqrt{3}$ m **(b)** $\frac{100}{\sqrt{3}}$ m **(c)** 50 m **(d)** 200 m

5. A student states: "The angle of depression of B from A is always greater than the angle of elevation of A from B." The correct response is: 1

(a) Correct
(b) Incorrect — they are equal (alternate angles)
(c) Correct only above 45°
(d) Cannot be determined

6. A kite string is 100 m long and makes 30° with the ground. The height of the kite is: 1

(a) 50 m **(b)** $50\sqrt{3}$ m **(c)** 100 m **(d)** $\frac{100}{\sqrt{3}}$ m

7. A 10 m tree casts a shadow $10\sqrt{3}$ m long. Student A says the sun's altitude is 30°; Student B says 60°. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. As the length of the shadow of a tower increases, the angle of elevation of the sun: 1

(a) increases **(b)** decreases **(c)** remains the same **(d)** becomes 90°

9. A bird sits on top of an 80 m tree; its angle of elevation from a point on the ground is 45°. The distance of the point from the tree is: 1

(a) 80 m **(b)** $80\sqrt{3}$ m **(c)** 40 m **(d)** $\frac{80}{\sqrt{3}}$ m

10. From a point 20 m from the foot of a tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) $20\sqrt{3}$ m **(b)** $\frac{20}{\sqrt{3}}$ m **(c)** 10 m **(d)** 20 m

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A tree 12 m tall casts a shadow $4\sqrt{3}$ m long. Find the angle of elevation of the sun. 2

12. From the top of a 10 m high building, the angle of depression of a point on the ground is 60°. Find the distance of the point from the foot of the building. 2

13. An electrician has to repair a fault on a 5 m high pole at a point 1.3 m below its top. His ladder is inclined at 60° to the horizontal. Find the length of the ladder needed. **2**

14. A 1.8 m tall man observes the top of a tower at 45° from a point 20 m away. He concludes that the tower is 20 m tall. Justify or refute. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. From a point on the ground 40 m away from the foot of a tower, the angle of elevation of the top of the tower is 30°. The angle of elevation of the top of a water tank on the tower is 45°. Find **3**
(i) the height of the tower
(ii) the depth of the tank.

16. A man in a boat rowing away from a 100 m high lighthouse takes 2 minutes to change the angle of elevation of the top of the lighthouse from 60° to 45°. Find the speed of the boat (in m/min). **3**

17. The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45°. Find the height of the multi-storeyed building and the distance between the two buildings. **3**

OR

The angle of elevation of the top of a hill from the foot of a tower is 60° and the angle of elevation of the top of the tower from the foot of the hill is 30°. If the tower is 50 m high, find the height of the hill. **3**

18. From a point P on the ground, the angle of elevation of the top of a 10 m tall building is 30°. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is 45°. Find the length of the flagstaff and the distance of the building from P. **3**

OR

A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank is 60°. When he moves 40 m away from the bank, the angle becomes 30°. Find the height of the tree and the width of the river. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. From the top of a 60 m high building, the angles of depression of the top and the bottom of a tower are 45° and 60° respectively. **5**
(i) Find the distance between the building and the tower.
(ii) Find the height of the tower.
(iii) A student says the tower is taller than the building. Comment.

OR

A man on the deck of a ship is 10 m above water level. He observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Find the distance of the hill from the ship and the height of the hill. **5**

20. **Case study:** A drone hovers at a height of 50 m above a point P of a playground. Two students A and B, on the same side of P and in line with it, observe the drone at angles of elevation 60° and 30°. **5**
(i) Find PA.
(ii) Find PB.
(iii) Find the distance AB.
(iv) Who is nearer to the drone?

OR

Case study: Two ships are sailing on either side of a lighthouse. The angles of depression of the ships from the top of the lighthouse are 60° and 45°. The distance between the ships is $\frac{100(\sqrt{3}+1)}{\sqrt{3}}$ m. **5**
(i) Express the distance of each ship in terms of the height h.
(ii) Find the height of the lighthouse.

END OF QUESTION PAPER — Total: 40 Marks